

## **Sprint 3(Information System 3)**

### **AI / Predictive Logic and AI Workflow Diagram**

#### **Introduction**

The SmartClinic Queue and Appointment Management System integrate predictive logic and smart workflow processing to improve patient flow, reduce waiting times, and assist clinic staff in decision-making. The system collects patient booking, appointment, and queue data from the frontend and backend modules, processes the information through the backend services, and generates queue predictions and occupancy monitoring results.

The predictive logic is designed to support:

- Queue monitoring
- Waiting-time estimation
- Occupancy tracking
- Appointment prioritisation
- Reporting and analytics

#### **AI / Predictive Logic**

##### **Front-End Data collection**

The frontend interface allows patients and clinic staff to:

- Book appointments
- Check in patients
- View queue information
- Update patient records

Input validation is implemented to ensure that required fields are completed correctly before data is submitted to the backend API.

##### **Backend Processing**

The backend API receives and processes appointments, queue, and sensor information. The system analyses:

- Number of active patients in the queue
- Appointment schedules
- Check-in timestamps
- Queue durations
- Occupancy levels in the waiting area

The backend then calculates estimated waiting times and identifies possible queue congestion periods.

### Predictive Logic Used

The system currently uses rule-based predictive logic and rolling-average calculations to estimate waiting times and queue conditions.

The predictive module:

- Calculates average waiting durations from queue data
- Predicts expected waiting time for patients
- Detects overcrowding based on occupancy thresholds
- Assists clinic staff in prioritising delayed or urgent appointments

This approach improves clinic efficiency and patient flow management.

### Data Storage and Analytics

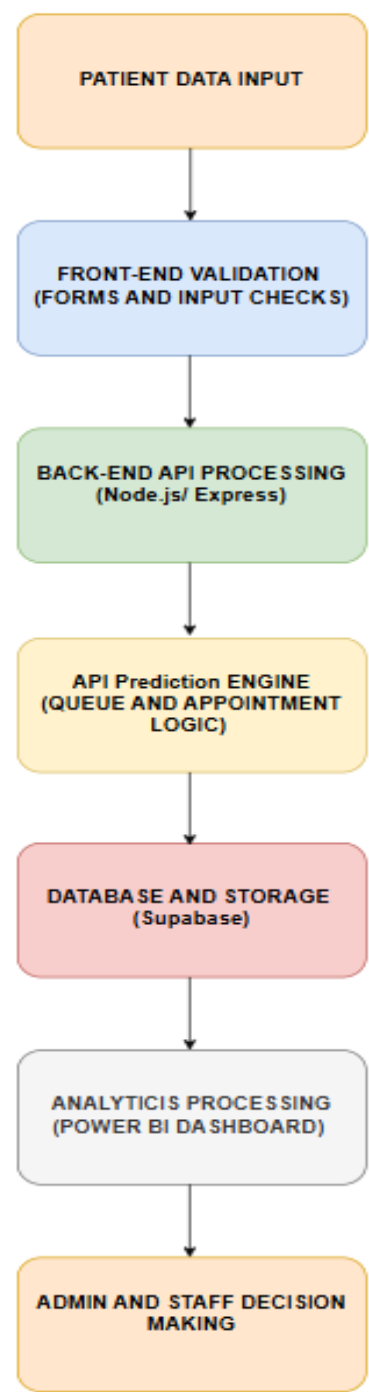
Processed queue and prediction data are stored in the project database through the backend services. The information can then be exported or connected to Power BI dashboards for:

- Queue trend analysis
- Patient statistics
- Clinic performance monitoring
- Occupancy reporting

### **AI Features Implemented**

| <b>AI / Smart Feature</b>  | <b>Purpose</b>                                                       |
|----------------------------|----------------------------------------------------------------------|
| Queue Prediction           | Predicts busy clinic periods using appointment and queue data        |
| Estimated Waiting Time     | Calculates expected patient waiting time using rolling-average logic |
| Occupancy Monitoring       | Tracks the number of patients currently active in the waiting area   |
| Appointment Prioritisation | Helps staff identify urgent or delayed appointments                  |
| Analytics Integration      | Supports Power BI reporting and visualisation                        |
| Input Validation           | Prevents incomplete or incorrect data submission                     |

**AI Workflow Diagram**



### Explanation of the Workflow

The workflow begins when a patient or clinic staff member enters appointment or check-in information through the frontend interface. The system validates the input data before sending it to the backend API.

The backend processes queue and appointment information and sends the data to the predictive logic module. The predictive logic analyses queue activity, occupancy levels, and average waiting durations to estimate patient waiting times and identify possible congestion.

The processed information is then stored in the database and used for reporting and analytics. Power BI dashboards can visualise queue trends, clinic occupancy, patient statistics, and overall clinic performance to support management decision-making.